Smart Campus Resource Booking System

Project Proposal for UCS503P

Submitted to: **Ms. Paramveer Sidhu**

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1 Introduction

Educational institutions have various resources such as classrooms, laboratories, seminar halls, projectors, sports facilities, and other equipment that require efficient management and scheduling. In many cases, the process of reserving these resources is handled manually or through disconnected systems, which can lead to scheduling conflicts, inefficient resource utilization, and difficulty in tracking reservations.

The **Smart Campus Resource Booking System** aims to provide a centralized web-based platform through which students, faculty members, and administrators can efficiently manage campus resources and their bookings.

The proposed system will allow users to view available resources, submit booking requests, track their bookings, and receive information regarding the status of their requests. Administrators will be able to manage resources, approve or reject booking requests, monitor utilization, and maintain the overall system.

This project is an extension of a **Resource Booking System developed as a DBMS project using Oracle**. The previous project primarily focused on database design, relationships, constraints, and SQL-based operations. The proposed UCS503 project will extend this foundation into a complete software application by introducing a user interface, application logic, authentication, booking workflows, validation, and software engineering practices.

2 Project Background

The initial Resource Booking System was developed as part of a Database Management System project. The system was implemented using Oracle Database and focused primarily on the storage and management of information related to users, resources, and bookings.

The existing database provides a foundation for the proposed system. Instead of developing the database component from the beginning, the existing design will be analyzed, improved where necessary, and integrated with the new application layer.

The major transition planned in this project is:

DBMS Project $\longrightarrow$ **Complete Web-Based Software System**

The new system will therefore demonstrate how an existing database-oriented solution can be evolved into a complete software product using systematic software engineering practices.

3 Problem Statement

Campus resources are shared among a large number of students, faculty members, and departments. Managing these resources manually can result in several operational problems.

The major problems include:

- Double booking of the same resource.
- Lack of centralized resource availability information.
- Difficulty in managing booking requests.
- Lack of proper booking history and tracking.
- Inefficient utilization of campus resources.
- Increased administrative workload.
- Difficulty in monitoring and generating resource usage reports.

A centralized software solution is therefore required to automate the resource booking process, reduce scheduling conflicts, and provide efficient management of campus resources.

4 Objectives

The primary objective of this project is to develop a web-based Smart Campus Resource Booking System that provides a centralized platform for managing and scheduling campus resources.

The specific objectives are:

- To develop a web-based system that allows authenticated users to browse resources, check availability, and submit booking requests.
- To implement automated availability and conflict validation to prevent overlapping bookings for the same resource and time slot.
- To provide administrators with functionality for managing resources, users, and booking requests.
- To maintain booking history and provide dashboards for monitoring current and previous reservations.
- To test and validate the system through functional, integration, and user acceptance testing.

5 System Architecture

The proposed system will follow a layered web application architecture consisting of the presentation layer, application layer, and database layer.

- **Presentation Layer:** Provides the web interface through HTML, CSS, JavaScript, and Bootstrap.

- **Application Layer:** Implements authentication, booking logic, resource management, validation, and administrator functionality using Java Servlets/JSP.
- **Database Layer:** Stores users, resources, bookings, availability, and related information using Oracle Database. JDBC will be used for communication between the Java application and the database.

The general interaction flow is:

User $\rightarrow$ Web Interface $\rightarrow$ Java Application $\rightarrow$ JDBC $\rightarrow$ Oracle Database

6 Proposed Solution

The proposed system will be developed as a web-based application that connects users with campus resources through a centralized booking platform.

The system will provide different functionalities according to the user's role.

6.1 User Functionality

Users will be able to:

- Register and log in to the system.
- Browse available campus resources.
- Search and filter resources.
- Check resource availability.
- Submit booking requests.
- Cancel eligible bookings.
- View current and previous bookings.
- Track the status of booking requests.

6.2 Administrator Functionality

Administrators will be able to:

- Log in through an administrator account.
- Add, update, and remove resources.

- Manage resource categories.
- Approve or reject booking requests.
- Manage user accounts.
- Monitor bookings.
- View resource utilization.
- Generate basic reports.

7 Scope of the Project

The project will focus on developing a functional campus resource booking system with the following major areas.

7.1 User Management

- User registration and authentication.
- Login and logout functionality.
- Role-based access control.
- User profile management.

7.2 Resource Management

- Adding and managing resources.
- Categorizing resources.
- Maintaining resource details.
- Tracking resource availability.

7.3 Booking Management

- Checking resource availability.
- Selecting date and time slots.
- Creating booking requests.
- Approving or rejecting booking requests.

- Preventing conflicting bookings.
- Maintaining booking history.

7.4 Dashboard and Reports

- User booking dashboard.
- Administrator dashboard.
- Upcoming booking information.
- Booking statistics.
- Resource utilization reports.

8 Major Modules

8.1 Authentication Module

This module will handle user registration, login, logout, and authorization. Access to different system functionalities will be controlled according to the user's role.

8.2 Resource Management Module

This module will manage the resources available within the campus. Administrators will be able to add, update, remove, and categorize resources.

8.3 Booking Module

The booking module will allow users to request resources for specific dates and time slots. The system will validate availability and prevent conflicting bookings.

8.4 Administration Module

The administration module will provide administrators with functionality for managing users, resources, booking requests, and system information.

8.5 Dashboard and Reporting Module

This module will provide summarized information about bookings and resource utilization. Administrators will be able to monitor booking activity and generate basic reports.

9 Technology Stack

The proposed technology stack is:

Category	Technology
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Java, Servlets/JSP, JDBC
Database	Oracle Database
Version Control	Git and GitHub
Development Environment	VS Code / IntelliJ IDEA
Documentation	LaTeX

10 Methodology

The project will follow an incremental software development approach in which the system will be developed, tested, evaluated, and improved in multiple iterations.

10.1 Requirement Analysis

The functional and non-functional requirements of the system will be identified. User roles, resource management requirements, booking workflows, and system constraints will be documented.

10.2 System Design

The system architecture, database structure, user interfaces, and major system workflows will be designed. UML diagrams such as use-case, activity, class, and sequence diagrams will be used where appropriate.

10.3 Database Analysis and Integration

The existing Oracle database developed during the previous DBMS project will be reviewed and refined where required. The database will then be integrated with the Java application using JDBC.

10.4 Application Development

The application will be developed incrementally. Authentication, resource management, booking management, administrator functionality, and dashboards will be implemented as separate functional modules.

10.5 Integration

The frontend, Java application layer, and Oracle database will be integrated to form a complete working system. Booking validation will be implemented to prevent conflicting reservations.

10.6 Testing and Validation

- **Functional Testing:** Verify individual system features such as login, resource search, booking, cancellation, and approval.
- **Integration Testing:** Verify communication between the frontend, Java application, and Oracle database.
- **Database Testing:** Verify data consistency, constraints, relationships, and correct storage of booking information.
- **Conflict Testing:** Verify that overlapping bookings for the same resource cannot be created.
- **User Acceptance Testing:** Evaluate whether the system meets the expected workflow and usability requirements.

10.7 Improvement and Deployment

Based on testing and evaluation feedback, identified issues will be resolved and the system will be refined before final deployment and presentation.

11 Project Timeline

The project will be developed incrementally throughout the semester. The proposed schedule is shown below.

12 Expected Outcomes

At the end of the project, the following outcomes are expected:

Weeks	Activities	Deliverable
W1–W2	Problem identification and requirement analysis	Requirements
W3	Project planning and elevator pitch	Project Concept
W4	Proposal and initial system design	Project Proposal
W5–W6	UML, database analysis and UI design	System Design
W7	Initial implementation	Prototype 1
W8	Prototype evaluation and improvement planning	Improvement Plan
W9–W10	Core application development	Functional Modules
W11–W12	System integration and testing	Integrated Prototype
W13	Buffer, debugging and improvements	Stabilized System
W14	Final feature refinement	Improved Prototype
W15	Second prototype and evaluation	Prototype 2
W16	Final testing and improvements	Final Candidate
W17	Final prototype, presentation and documentation	Final Project

- A functional web-based campus resource booking system.
- A centralized platform for managing campus resources.
- Reduced chances of double booking and scheduling conflicts.
- Improved visibility of resource availability.
- Efficient management of booking requests.
- Booking history and resource utilization information.
- Role-based access for users and administrators.
- Complete project documentation following software engineering practices.

13 Future Enhancements

The following features may be considered as future enhancements depending on the available development time:

- Email notifications for booking status and reminders.
- QR-based booking verification.
- Advanced resource utilization analytics.
- Mobile application support.
- Automated maintenance scheduling for resources.
- AI-based prediction of resource demand.

14 Risks and Mitigation

The following risks have been identified during the initial planning of the project:

- **Learning Curve:** Developing the Java web application may require learning new technologies. This will be addressed through incremental development, beginning with Java, JDBC, and basic web application concepts.
- **Database Integration:** The existing Oracle database may require modifications during application integration. The database schema will therefore be reviewed and tested before integration.
- **Limited Development Time:** The project must be completed alongside regular coursework. Core booking functionality will therefore be prioritized before optional features.
- **Scope Expansion:** Additional features may increase development complexity. Optional features will only be implemented after the core system is functional.

15 Team Details

Name	**Primary Responsibility**
Dhruv Bihani	Backend Development and Database Integration
Aditya Srivastava	Frontend Development and Testing

Both team members will collaboratively participate in requirement analysis, system design, documentation, GitHub management, testing, presentations, and final deployment.